Compensating Baseband Distortion of Regularly Sampled Pulsewidth Modulators for High-Precision Power Converters

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***Abstract*—This paper presents a straight-forward method to compensate baseband distortion for regularly sampled pulsewidth modulation, for both sawtooth and triangular carrier, aimed at high-precision power electronics applications such as medical MRI and semiconductor lithography. The method adds correction signals to the modulator input to mitigate the distortion, which is computationally lightweight, and pseudo-code is given to simplify implementation. Measurements are conducted, which confirm the improvements offered by the method. It is shown that baseband distortion can be reduced by upto 60 dB. This improves the linearity and reduces audible noise of power converters.**

***Index Terms*—Nonlinear distortion, power quality, predistortion, pulse width modulation converters.**

I. Introduction

PULSEWIDTH modulation (PWM) is commonly used to construct the desired output voltage in switched-mode power amplifiers. Either digital or analog implementations can be used, often referred to as regularly sampled and natural-sampled PWM respectively. Regularly sampled PWM is nowadays the preferred choice since most applications employ digital control loops.

In most applications of high-precision power electronics, such as gradient amplifiers in medical MRI and wafer positioning systems in semiconductor lithography, there is a continuous strive toward higher bandwidth and higher precision. However, the regularly sampled PWM that is often used in these applications suffers from a non-linear input–output relation that causes baseband distortion [1]–[3], and thereby limits the precision that can be achieved. When power conversion topologies with very low output distortion are used, such as the dual-buck converter [4], [5], this baseband distortion becomes a limiting factor for further improving the accuracy. This is especially the case when the ratio between the carrier frequency and the reference frequency is relatively low (i.e., high bandwidth). For example, with a ratio of 20, the total harmonic distortion (THD) for a symmetrically sampled triangular-carrier PWM is still high, namely $\mathrm{THD}_7 \approx 3.9\%$ (taking the first seven harmonics into account). To reduce the distortion, this ratio can be increased. However, there is a limit to which the carrier frequency can be increased, since this sacrifices the PWM resolution (thus adds quantization noise), as well as increases the switching losses in the power semiconductors.

There are a few publications on reducing this type of distortion by compensation. An exact solution for this problem is provided by [6] and [7]; however, the application of such algorithm is complex, and real-time implementation requires significant processing power. Another method that is commonly used is pseudo-natural PWM [8], [9], which operates by predicting the reference-carrier intersections to mimic natural sampling.

This paper presents an approximation that provides good results and is easy to implement in a microcontroller or digital signal processor (DSP) for real-time use. Pseudo-code is given to ease the implementation of the compensator. The compensator structure is similar to what is presented in [10] and [11]. However, it must be noted that their solutions are not consistent, and the coefficients differ from what is derived in this paper as well. Moreover, this paper includes more carrier types and a more extensive derivation of the coefficients. Furthermore, pseudo-code is given to ease the implementation. Validation is done by simulations and measurements to ensure the correctness of the solutions.

The structure of this paper is as follows. First, the used modeling and compensation technique is presented. Then, the modulator models are derived and the compensators are determined for regularly sampled sawtooth-carrier and triangular-carrier PWMs, with symmetrical sampling and asymmetrical sampling. Next, pseudo-code for ease of implementation is given. Finally, simulation and measurement results are shown and conclusions are given.

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II. Distortion Model and Compensation Filter

The model of the PWM and its compensation filter are determined for regularly sampled PWM. For this, a nonlinear mathematical structure is used that is known as a Hammerstein/Wiener-model [12], which enables the modeling of non-linear systems. The model of the pulsewidth modulator

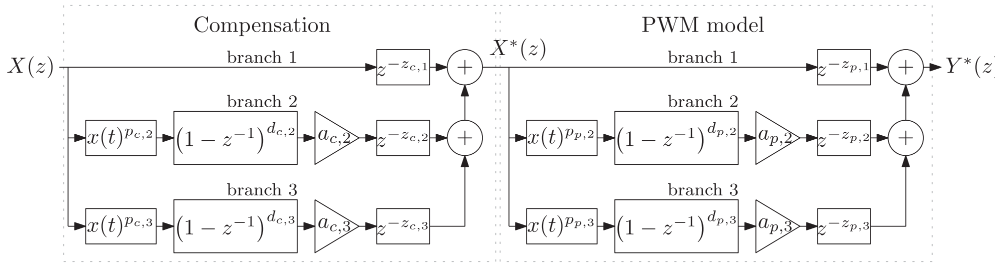

Fig. 1. Structure as used for baseband distortion compensation and modeling the pulsewidth modulator.

is fitted onto the non-linear Z-domain equation given by

$$Y_p(z)=X(z)\cdot z^{-z_{p,1}}+\sum_{k=2}^{k_p}a_{p,k}\mathcal{Z}(x(t)^{p_{p,k}})(1-z^{-1})^{d_{p,k}}z^{-z_{p,k}}. \tag{1}$$

Here, $Y_p(z)$ is the PWM model output, $X(z)$ is the Z-transformed PWM input signal, $z^{-z_{p,k}}$ is a delay of $z_{p,k}$ samples, $a_{p,k}$ is the linear gain, $\mathcal{Z}\left(x(t)^{p_{p,k}}\right)$ is the Z-transform of $x(t)^{p_{p,k}}$, and $\left(1-z^{-1}\right)^{d_{p,k}}$ is the $d_{p,k}$th discrete derivative for model branch k. The compensated modulator-model output is labeled $Y_p^*(z)$ with input $X^*(z)$. Its compensation filter is fitted onto a similar model, being

$$X(z)^*=X(z)\cdot z^{-z_{c,1}}+\sum_{k=2}^{k_c}a_{c,k}\mathcal{Z}(x(t)^{p_{c,k}})(1-z^{-1})^{d_{c,k}}z^{-z_{c,k}}. \tag{2}$$

In the presented approach, each branch k is used for modeling or compensating one harmonic component. It is chosen that both the model and the compensation filter are determined for three branches, i.e., $k_c = 3$ and $k_p = 3$, which allows modeling and compensating the first two harmonics adequately. The overall structure is graphically represented in Fig. 1.

Throughout the analysis, some approximations are used and the conditions under which these are accurate within 1% are stated. The value of 1% is chosen since the improvement in distortion is then 40 dB (upper bound). It is assumed that the original uncompensated reference is equal to

$$x(t) = M\cos(\omega_o t). \tag{3}$$

Moreover, it is assumed that the carrier amplitude is unity [−1, 1]. The Bessel-function of the first kind is approximated by its asymptotic form

$$J_n(x) \approx \frac{1}{n!}\left(\frac{x}{2}\right)^n \tag{4}$$

which is accurate when $x \ll 1$.

The non-linear response of (1) must match with the actual PWM response. To do this, the output of model branch k is transformed to the time domain with n_s samples per carrier period, being

$$y_{p,k}(t) = a_{p,k}\left(\frac{2\pi}{n_s\omega_c}\right)^{d_{p,k}}\frac{d^{d_{p,k}}}{dt^{d_{p,k}}}\left(x(t)^{p_{p,k}}\right). \tag{5}$$

It can be seen that each branch generates sub-harmonics when $P_{p.n} <> 1$. Since in general $Y_{n+1} \ll Y_n$, the lower-order branches will dominate the sub-harmonics of the higher-order branches, such that only the highest frequency generated in each branch needs to be considered. This results in the approximated contribution for each branch

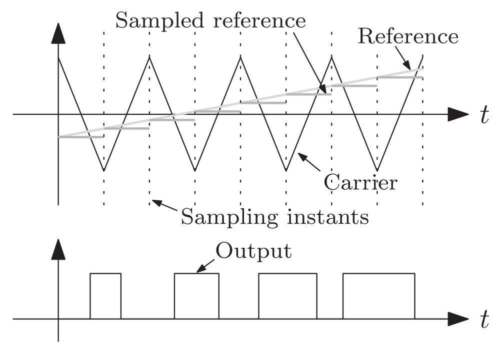

Fig. 2. Modulator waveforms with triangular carrier and asymmetric sampling.

$$y_{p,k}(t) \approx \frac{a_{p,k}M^{p_{p,k}}}{2^{p_{p,k}-1}}\left(\frac{2\pi}{n_s\omega_c}\right)^{d_{p,k}}\frac{d^{d_{p,k}}}{dt^{d_{p,k}}}\left(\cos\left(p_{p,k}\omega_o t\right)\right) \tag{6}$$

such that the relevant harmonic magnitude of each branch equals

$$Y_{p,k} = \frac{a_{p,k}\left(2\pi p_{p,k}\right)^{d_{p,k}}}{2^{p_{p,k}-1}\, n_s^{d_{p,k}}}\left(\frac{\omega_o}{\omega_c}\right)^{d_{p,k}} \times M^{p_{p,k}}\left(\cos\left(d_{p,k}\frac{\pi}{2}\right)-j\sin\left(d_{p,k}\frac{\pi}{2}\right)\right). \tag{7}$$

The same method can be applied to (2).

III. Triangular Carrier—Asymmetric Sampling

The modulator with a triangular carrier and asymmetric sampling is modeled according to the waveforms shown in Fig. 2. The reference is sampled twice per period, at the center of the output OFF and ON states.

A. PWM Model

The baseband harmonics are calculated using the method proposed in [1], and it is shown that the baseband harmonics with asymmetric sampling and a triangular carrier are equal to

$$Y_n = \frac{4}{n\pi\left(\frac{\omega_o}{\omega_c}\right)}J_n\left(n\left(\frac{\omega_o}{\omega_c}\right)\left(\frac{\pi}{2}\right)M\right)\sin\left(n\frac{\pi}{2}\right). \tag{8}$$

It must be noted that only the PWM-process is considered, and not the overall power converter, hence V_{dc} is removed from the equations from [1]. Applying (4) results in the approximation

$$Y_n \approx \frac{(n\pi)^{n-1}}{4^{n-1}n!}\left(\frac{\omega_o}{\omega_c}\right)^{n-1} M^n \sin\left(n\frac{\pi}{2}\right) \tag{9}$$

which is accurate up to 1% for a minimum ratio of $\frac{\omega_c}{\omega_o} > 16$ for all valid M. Note that the baseband components are purely real, meaning only cosine-components are present. Then, the model coefficients are found by letting $Y_{p,n} = Y_n$ with two samples per period ($n_s = 2$), such that

$$\begin{aligned}\frac{(n\pi)^{n-1}}{4^{n-1}n!}\left(\frac{\omega_o}{\omega_c}\right)^{n-1} M^n \sin\left(n\frac{\pi}{2}\right) &= \frac{a_{p,k}\,(\pi p_{p,k})^{d_{p,k}}}{2^{p_{p,k}-1}}\left(\frac{\omega_o}{\omega_c}\right)^{d_{p,k}} M^{p_{p,k}}\ \left(\cos\left(d_{p,k}\frac{\pi}{2}\right)\right.\\ &\quad \left.-j\sin\left(d_{p,k}\frac{\pi}{2}\right)\right).\end{aligned} \tag{10}$$

It can now be seen that for this modulator type, it holds that $d_{p,k} = n-1$ and $p_{p,k} = n$, with $n = \{3, 5, 7, \ldots\}$, $k = \{2, 3, 4, \ldots\}$. The gain coefficients are then equal to

$$a_{p,k} = \frac{2^{n-1}}{4^{n-1}n!}. \tag{11}$$

The second branch ($k = 2$) is used for compensating the third harmonic ($n = 3$), then the coefficients for this branch are $d_{p,2} = 2$, $p_{p,2} = 3$. The third branch ($k = 3$) is used for compensating the fifth harmonic ($n = 5$), then the coefficients for this branch are $d_{p,3} = 4$, $p_{p,3} = 5$. The corresponding gain coefficients of the PWM model are equal to

$$a_{p,2} = \frac{1}{24} \quad \text{and} \quad a_{p,3} = \frac{1}{1920}. \tag{12}$$

Discrete differentiation has a time delay of half the sample period time. This requires a delay element in the branches to ensure time alignment. For each differentiator, a delay of half the sample period time must be added to the other branches. Hence

$$z_{p,1} = \frac{\max\left(d_{p,k}\,|\,\{n \in [1\ldots k_p]\}\right)}{2} = 2 \tag{13}$$

$$z_{p,2} = \frac{\max\left(d_{p,k}\,|\,\{n \in [1\ldots k_p]\}\right)}{2} - \frac{d_{p,2}}{2} = 1 \tag{14}$$

$$z_{p,3} = \frac{\max\left(d_{p,k}\,|\,\{n \in [1\ldots k_c]\}\right)}{2} - \frac{d_{p,3}}{2} = 0. \tag{15}$$

B. Compensation Filter

The compensation filter is calculated by using the model derived in Section III-A as follows. First, the reference signal is selected according to (3). Next, a three-branch compensation filter is defined according to (2), with $d_{c,2} = d_{p,2}$, $d_{c,3} = d_{p,3}$, $p_{c,2} = p_{p,2}$, and $p_{c,3} = p_{p,3}$ to match the frequency and modulation index dependency of the model and the compensation filter. The resulting PWM input $x^*(t)$ is then fed into (1) to calculate the time-domain response of $y^*(t)$ using (5), taking into account all sub-harmonics of each branch. Assuming $\left(\frac{\omega_o}{\omega_c}\right)^k \gg \left(\frac{\omega_o}{\omega_c}\right)^{k+1}$, which holds when $\frac{\omega_c}{\omega_o} \to \infty$, a reduced-order model can be constructed, being for the first, second, and third harmonic

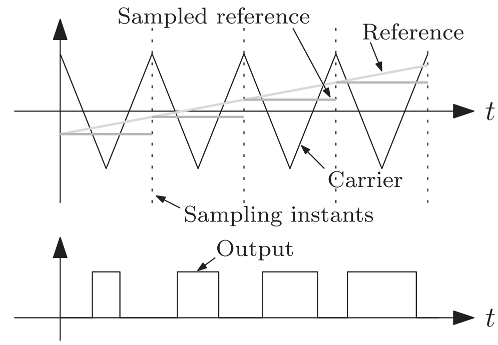

Fig. 3. Modulator waveforms with triangular carrier and symmetric sampling.

equal to

$$\begin{aligned} y^*_{\text{red}}(t) = M\cos(\omega_o t) &- \frac{9a_{c,2}M^3\pi^2\cos(3\omega_o t)\,\omega_o^2}{4\omega_c^2}\\ &- \frac{9a_{p,2}M^3\pi^2\cos(3\omega_o t)\,\omega_o^2}{4\omega_c^2}\\ &+ \frac{625a_{c,3}M^5\pi^4\cos(5\omega_o t)\,\omega_o^4}{16\omega_c^4}\\ &+ \frac{675a_{c,2}a_{p,2}M^5\pi^4\cos(5\omega_o t)\,\omega_o^4}{16\omega_c^4}\\ &+ \frac{625a_{p,3}M^5\pi^4\cos(5\omega_o t)\,\omega_o^4}{16\omega_c^4}. \end{aligned} \tag{16}$$

To allow $y^*_{\text{red}}(t) \to x(t)$, the harmonic components must cancel out. This means

$$-a_{c,2} - a_{p,2} = 0 \ \text{and} \ 625a_{c,3} + 675a_{c,2}a_{p,2} + 625a_{p,3} = 0 \tag{17}$$

resulting in the coefficient values

$$a_{c,2} = \frac{-1}{24} \quad \text{and} \quad a_{c,3} = \frac{13}{9600}. \tag{18}$$

Similarly to the PWM model, a delay element must be inserted in the branches to ensure time alignment. For each differentiator, a delay of the sample period time must be added to the other branches. Hence

$$z_{c,1} = \frac{\max\left(d_{c,k}\,|\,\{n \in [1\ldots k_c]\}\right)}{2} = 2 \tag{19}$$

$$z_{c,2} = \frac{\max\left(d_{c,k}\,|\,\{n \in [1\ldots k_c]\}\right) - d_{c,2}}{2} = 1 \tag{20}$$

$$z_{c,3} = \frac{\max\left(d_{c,k}\,|\,\{n \in [1\ldots k_c]\}\right) - d_{c,3}}{2} = 0. \tag{21}$$

IV. Triangular Carrier—Symmetric Sampling

The modulator with a triangular carrier using symmetric sampling is modeled according to the waveforms shown in Fig. 3. Here, the reference is sampled once per period, at the center of the output OFF state.

A. PWM Model

The baseband harmonics for a triangular carrier with symmetric sampling are calculated using the method proposed in [1], and are found to be

$$Y_n = \frac{4}{n\pi\left(\frac{\omega_o}{\omega_c}\right)} J_n\left(n\left(\frac{\omega_o}{\omega_c}\right)\left(\frac{\pi}{2}\right)M\right)\sin\left(n\left(1+\frac{\omega_o}{\omega_c}\right)\frac{\pi}{2}\right). \tag{22}$$

Applying (4), and approximating $\sin(n(1+\frac{\omega_o}{\omega_c})\frac{\pi}{2}) \approx \sin(n\frac{\pi}{2}) + (\frac{\omega_o}{\omega_c})(n\frac{\pi}{2})\cos(n\frac{\pi}{2})$, results in

$$Y_n \approx \frac{(n\pi)^{n-1}}{4^{n-1}n!}\left(\frac{\omega_o}{\omega_c}\right)^{n-1} M^n \left(\sin\left(n\frac{\pi}{2}\right) + \left(\frac{\omega_o}{\omega_c}\right)\left(n\frac{\pi}{2}\right)\cos\left(n\frac{\pi}{2}\right)\right). \tag{23}$$

Then, equally to the method discussed in Section III-A, the PWM model is constructed using (1). After manipulations with one sample per period ($n_s = 1$), it is found that

$$\begin{aligned}&\frac{(n\pi)^{n-1}}{4^{n-1}n!}\left(\frac{\omega_o}{\omega_c}\right)^{n-1} M^n \left(\sin\left(n\frac{\pi}{2}\right) + \left(\frac{\omega_o}{\omega_c}\right)\left(n\frac{\pi}{2}\right)\cos\left(n\frac{\pi}{2}\right)\right)\\ &= \frac{a_{p,k}\,(2\pi p_{p,k})^{d_{p,k}}}{2^{p_{p,k}-1}}\left(\frac{\omega_o}{\omega_c}\right)^{d_{p,k}}\\ &\quad\times M^{p_{p,k}}\left(\cos\left(d_{p,k}\frac{\pi}{2}\right) - j\sin\left(d_{p,k}\frac{\pi}{2}\right)\right)\end{aligned} \tag{24}$$

such that $d_{p,k} = n$ and $p_{p,k} = n$ for n is even, and $d_{p,k} = (n-1)$ and $p_{p,k} = n$ for n is odd.

The second branch $k = 2$ is used for the second harmonic $n = 2$, and the third branch $k = 3$ is used for the third harmonic $n = 3$, and it is then found that $d_{p,2} = 2$, $d_{p,3} = 2$, $p_{p,2} = 2$, and $p_{p,3} = 3$. The gain coefficients are equal to

$$a_{p,k} = \begin{cases}\dfrac{1}{4^n n!}, & n \text{ even}\\[2mm] \dfrac{1}{4^{n-1}n!}, & n \text{ odd}\end{cases} \tag{25}$$

resulting in

$$a_{p,2} = -\frac{1}{32} \quad a_{p,3} = \frac{1}{96}. \tag{26}$$

Time alignment is achieved by letting

$$z_{p,1} = \frac{\max\left(d_{p,k}\,|\,\{n \in [1\ldots k_p]\}\right)}{2} = 1 \tag{27}$$

$$z_{p,2} = \frac{\max\left(d_{p,k}\,|\,\{n \in [1\ldots k_p]\}\right) - d_{p,2}}{2} = 0 \tag{28}$$

$$z_{p,3} = \frac{\max\left(d_{p,k}\,|\,\{n \in [1\ldots k_p]\}\right) - d_{p,3}}{2} = 0. \tag{29}$$

B. Compensation Filter

The compensation filter is calculated by using the model derived in Section IV-A and in a fashion similar to Section III-B, with $d_{c,2} = d_{p,2}$, $d_{c,3} = d_{p,3}$, $p_{c,2} = p_{p,2}$, and $p_{c,3} = p_{p,3}$ to match the frequency and modulation index dependency of the model and the compensation filter. The coefficients are found using a reduced-order model and are equal to

$$a_{c,2} = -a_{p,2} = \frac{1}{32} \quad \text{and} \quad a_{c,3} = -a_{p,3} = -\frac{1}{96}. \tag{30}$$

Finally, proper time alignment of the discrete-time differentiators is achieved by letting

$$z_{c,1} = \max\left(d_{c,k}\,|\,\{n \in [1\ldots k_c]\}\right) = 1 \tag{31}$$

$$z_{c,2} = \max\left(d_{c,k}\,|\,\{n \in [1\ldots k_p]\}\right) - d_{c,2} = 0 \tag{32}$$

$$z_{c,3} = \max\left(d_{c,k}\,|\,\{n \in [1\ldots k_p]\}\right) - d_{c,3} = 0. \tag{33}$$

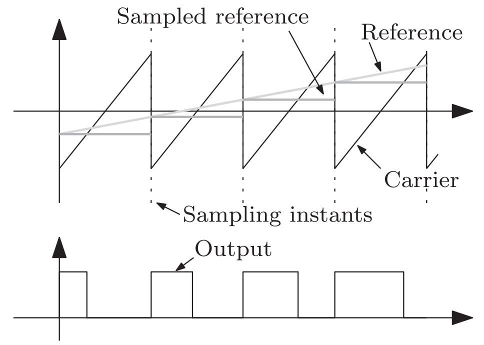

Fig. 4. Modulator waveforms with sawtooth carrier and regular sampling.

V. Sawtooth Carrier

The modulator with a sawtooth carrier and regular sampling is modeled according to the waveforms shown in Fig. 4. The reference is sampled once per period, at the moment the output transitions to ON.

A. PWM Model

The baseband harmonics for a sawtooth carrier with regular sampling are calculated using the method from [1], and are found to be

$$Y_n = \frac{2}{n\pi\left(\frac{\omega_o}{\omega_c}\right)} J_n\left(n\left(\frac{\omega_o}{\omega_c}\right)\pi M\right)\left(\sin\left(n\frac{\pi}{2}\right) - j\cos\left(n\frac{\pi}{2}\right)\right). \tag{34}$$

Applying (4) results in

$$Y_n \approx \frac{(n\pi)^{n-1}}{2^{n-1}n!}\left(\frac{\omega_o}{\omega_c}\right)^{n-1} M^n \left(\sin\left(n\frac{\pi}{2}\right) - j\cos\left(n\frac{\pi}{2}\right)\right). \tag{35}$$

Then, equally to the method discussed in Section III-A, the PWM model is constructed using (1). After manipulations with

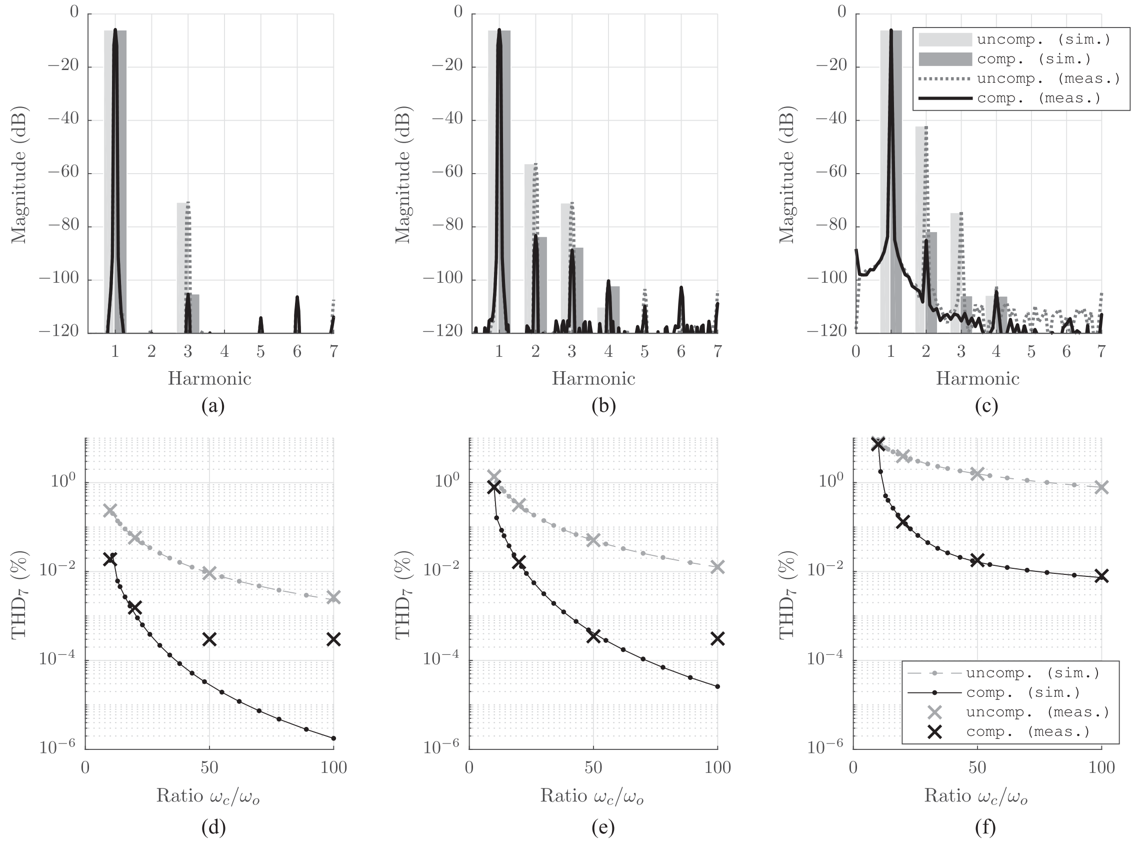

Fig. 5. Comparison of baseband distortion for uncompensated and compensated PWM using simulations and measurements. Left to right: asymmetric triangular, symmetric triangular, sawtooth. (a)–(c) Spectra for $\omega_c/\omega_o = 20$. (d)–(f) THD for various ω_c/ω_o ratios.

one sample per period ($n_s = 1$), it is found that

$$\frac{(n\pi)^{n-1}}{2^{n-1}n!}\left(\frac{\omega_o}{\omega_c}\right)^{n-1} M^n \left(\sin\left(n\frac{\pi}{2}\right) - j\cos\left(n\frac{\pi}{2}\right)\right)$$
$$= \frac{a_{p,k}\,(2\pi p_{p,k})^{d_{p,k}}}{2^{p_{p,k}-1}}\left(\frac{\omega_o}{\omega_c}\right)^{d_{p,k}}$$
$$\times M^{p_{p,k}}\left(\cos\left(d_{p,k}\frac{\pi}{2}\right) - j\sin\left(d_{p,k}\frac{\pi}{2}\right)\right) \quad (36)$$

such that $d_{p,k} = n-1$ and $p_{p,k} = n$. The second branch $k = 2$ is used for the second harmonic $n = 2$, and the third branch $k = 3$ is used for the third harmonic $n = 3$, and it is then found that $d_{p,2} = 1$, $d_{p,3} = 2$, $p_{p,2} = 2$, and $p_{p,3} = 3$. The gain coefficients are then equal to

$$a_{p,k} = \begin{cases} \dfrac{-1}{2^{n-1}n!}, & n \text{ even} \\ \dfrac{1}{2^{n-1}n!}, & n \text{ odd} \end{cases} \quad (37)$$

resulting in

$$a_{p,2} = -\frac{1}{4} \quad \text{and} \quad a_{p,3} = \frac{1}{24}. \quad (38)$$

Time alignment is achieved by letting

$$z_{p,1} = \frac{\max\left(d_{p,k}\,|\,\{n \in [1\ldots k_p]\}\right)}{2} = 1 \quad (39)$$

$$z_{p,2} = \frac{\max\left(d_{p,k}\,|\,\{n \in [1\ldots k_p]\}\right) - d_{p,2}}{2} = 0.5 \quad (40)$$

$$z_{p,3} = \frac{\max\left(d_{p,k}\,|\,\{n \in [1\ldots k_p]\}\right) - d_{p,3}}{2} = 0. \quad (41)$$

It can be seen that the second branch requires half a sample delay, for which a fractional-order delay is used. This can be implemented by using

$$z^{-1/2} \approx \frac{c^3 + 3cz^{-1}}{3c^2 + z^{-1}} \quad (42)$$

where $c = 1.4$ for minimum approximation error [13].

B. Compensation Filter

The compensation filter is found by using the model derived in the previous section, with $d_{c,2} = d_{p,2}$, $d_{c,3} = d_{p,3}$, $p_{c,2} = p_{p,2}$, and $p_{c,3} = p_{p,3}$ to match the frequency and modulation index dependency of the model and the compensation filter. A reduced-order model is then used to calculate the gain coefficients, which are found to be

$$a_{c,2} = -a_{p,2} = \frac{1}{4} \quad \text{and} \quad a_{c,3} = \frac{12}{9}a_{p,2}^2 - a_{p,3} = \frac{1}{24}. \quad (43)$$

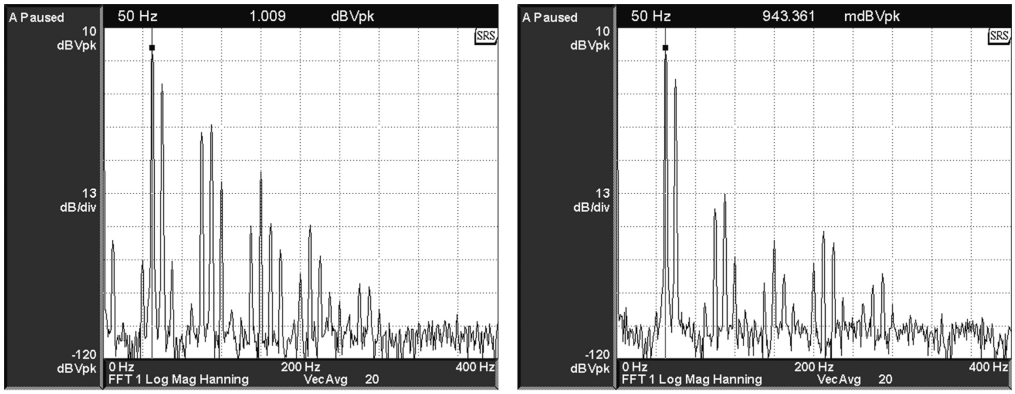

Fig. 6. Comparison of spectra for two-tone test with sawtooth-carrier PWM. Input signals are 50 Hz, M = 0.5 and 60 Hz, M = 0.25. (a) Uncompensated. (b) Compensated modulator.

Finally, time alignment of the discrete-time differentiators is achieved by letting

$$z_{p,1} = \frac{\max\left(d_{p,k} \,|\, \{n \in [1 \ldots k_p]\}\right)}{2} = 1 \tag{44}$$

$$z_{p,2} = \frac{\max\left(d_{p,k} \,|\, \{n \in [1 \ldots k_p]\}\right) - d_{p,2}}{2} = 0.5 \tag{45}$$

$$z_{p,3} = \frac{\max\left(d_{p,k} \,|\, \{n \in [1 \ldots k_p]\}\right) - d_{p,3}}{2} = 0. \tag{46}$$

Also for the compensator, the second branch requires half a sample delay, which can be approximated by using (42).

VI. Implementation

The compensator can be implemented on a DSP into a function call that finds the corrected PWM input signal at each sample interval. In Listing 1, an example implementation using pseudo-code is given. As can be seen, this calculation requires relatively little computational effort and it can be implemented to run in real time on a DSP. The time alignment of the first branch, $z_{c,1}$, determines the delay caused by the compensation scheme, which is equal to one or two samples depending on the carrier and sampling scheme. The delay can be reduced by using two instead of three branches, thereby compensating only one baseband harmonic.

VII. Results

To validate the presented method, both simulations and measurements are done. Simulations are carried out using MATLAB Simulink, while measurements are conducted on an experimental setup as shown in Fig. 7. The setup consists of a real-time dSPACE Microlabbox platform, a Stanford Research Systems spectrum analyzer, and a control PC. The measurements are carried out using a PWM frequency of 1 kHz, and a fundamental frequency of 50 Hz (unless noted otherwise). The PWM frequency is chosen low and noise shaping is implemented, both with the aim to reduce quantization noise as much as possible. The compensator runs in real time, synchronous with the PWM sampling.

```
function [x*]=CorrectionFilter(x)
        n=n+1; x_v[n]=x;
 %% step 1: power, differentiation, gain
        x*_{1,1}[n]=x_v[n];
        x*_{1,2}[n]=a_{c,2}*Diff(x_v^p_{c,2},n,d_{c,2});
        x*_{1,3}[n]=a_{c,3}*Diff(x_v^p_{c,3},n,d_{c,3});
 %% step 2: time alignment
        x*_{2,1}[n]=Delay(x*_{1,1},x*_{2,1},n,z_{c,1});
        x*_{2,2}[n]=Delay(x*_{1,2},x*_{2,2},n,z_{c,2});
        x*_{2,3}[n]=Delay(x*_{1,3},x*_{2,3},n,z_{c,3});
 %% step 3: composition of full signal
        x*=x*_{2,1}[n]+x*_{2,2}[n]+x*_{2,3}[n];
end
function [y]=Diff(x,n,d)
        k=0:d;
        y=sum(nchoosek(d,k)*x[n-k]*cos(k*pi));
end
function [y]=Delay(x,y,n,z)
        if(z==0.5)
                y=0.4667*x[n]+0.7143*x[n-1]-
                        0.1701*y[n-1];
        else
                y=x[n-z];
        end
end
```

Listing 1: Pseudo-code for DSP implementation of baseband distortion compensation.

Fig. 7. Experimental setup for distortion measurements of power converter. From left to right: oscilloscope, dSPACE MicroLabBox, control PC, Stanford Research Systems SR780 spectrum analyzer.

First, the modulator output spectrum is measured for a ratio $\omega_c/\omega_o = 20$ and $M = 0.5$, using uncompensated PWM and the proposed compensation method. The results are shown in Fig. 5(a)–(c). As can be seen, a significant distortion reduction

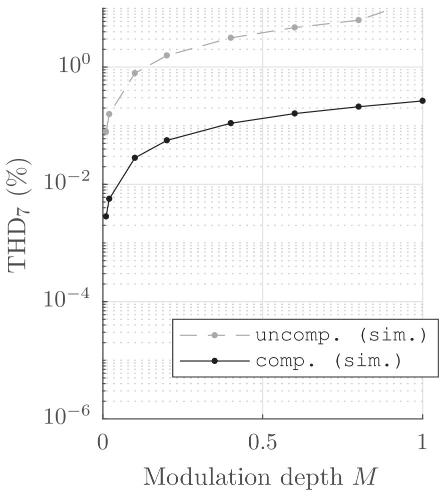

Fig. 8. Influence of modulation depth M on THD_7 with triangular carrier and $\omega_c/\omega_o = 20$.

is realized in the baseband and there is a good agreement with the simulation results.

Second, the THD_7 of the modulator output is simulated and measured for a wide range of ω_c/ω_o-ratios, using uncompensated PWM and the proposed compensation method. The results are depicted in Fig. 5(d)–(f), and it can be seen that a significant reduction is already achieved at very low ratios, and further improves to up to 60 dB for higher ratios. For example, at a ratio of 20, the THD_7 for symmetric sampled PWM improves from 0.31% to 0.016%, for the uncompensated and the compensated PWM, respectively. At a higher ratio of 50, the THD_7 improves from 0.051% to 0.00035%. It can be seen that measurements and simulations are in good agreement with each other for distortion levels above 0.0005%. The measurement equipment has a physical limitation at around 0.00035% THD where the output distortion of the converter and the spectrum analyzer accuracy is a limiting factor rather than the PWM baseband distortion, which can be clearly seen in the THD results.

Then, a two-tone measurement is conducted on the sawtooth carrier PWM, without and with compensation. These results are depicted in Fig. 6(a) and (b), respectively. It can be seen that the improvement in baseband distortion also applies for the two-tone test, and that inter-modulation distortion is reduced as well.

Finally, the modulation depth M is varied from 0.01 up to 1 for ratio $\omega_c/\omega_o = 20$ with a sawtooth-carrier PWM. It can be seen in Fig. 8 that the baseband distortion increases for increasing M, and that the improvement factor is approximately constant for all M.

VIII. Conclusion

This paper describes a non-linear model for regularly sampled pulsewidth modulators using a triangular carrier, for power electronic converters. Moreover, a method is presented to compensate the baseband distortion that is caused by these non-linearities. Simulations and measurements are conducted to verify the proposed compensation scheme, and an example implementation is given for easy application of the algorithm.

The theoretical distortion can be reduced by up to 60 dB, which significantly improves the THD of the modulator. The results are shown to be accurate within the tolerances of the measurement equipment. Therefore, by using the baseband distortion compensation method presented in this paper, it is possible to use a lower PWM frequency for the same distortion levels. This reduces the switching losses. Moreover, the lower distortion improves performance in applications such as gradient amplifiers in MRI and wafer positioning systems in semiconductor lithography. Compliance with harmonics standards in grid-connected applications is also simplified, and audible noise is reduced in PV applications, as well as motor drive and robotics applications.

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